Virginia Tech
Department of Economics
Spring 2026

ECON 3254: Applied Econometrics[1]

Instructor: Ph.D. [c] Martin A. Vargas Alvines (mavargasalvines@vt.edu)

Class Time: Tuesdays and Thursdays
2:00 p.m. – 3:15 pm (Section 21066)

Class Location: SEITZ 207 (Seitz Hall)

Instructor Office Hour Times: Tuesdays: 11 a.m. – 1:00 p.m. and
Wednesdays: 1:30 p.m. – 3:30 p.m.
3055, Pamplin Hall (3rd floor)
Or by appointment (email)

Course Description and Objectives

This course intends to expose you to the statistical techniques that economists use for estimating, testing, and forecasting economic relationships. The emphasis is on understanding the techniques involved and what they mean in terms of the economic problem being studied. Successful completion of this course should allow you to (1) understand empirical literature in economics and (2) be prepared to start doing independent research using economic data, particularly in upper-level classes. This course will emphasize *economic applications over statistical theory*.

Through this course, you will have gained the skills necessary to be a thoughtful consumer of social science research. You will be able to read and understand the output that is generated when you estimate a model. You will be able to identify and analyze the basic problems that can arise when using ordinary least squares estimation.

[1] Important: You should carefully read all information given in this syllabus. If you want to email me with a question, you should check the syllabus to see if your question is answered here before asking me. As a course policy, I direct students to the syllabus for questions already covered in that document.

Students often find the course to be rigorous and demanding. **We will move quickly and cover a large amount of material**. I advise you to prepare for class and keep up with the material, as the course does not lend itself to memorization or all-night cram sessions.

Prerequisites
BIT 2405 or MSCI 2405, BIT 2406, or MSCI 2406, or STAT 3005, or STAT 4604, or STAT 4705, or STAT 4714. For all these courses, MIN grade of P.

Course Materials

1. "Data Analysis for Business, Economics, and Policy", 2021, Békés, G., and Kézdi, G.
2. "Mastering 'Metrics: The Path from Cause to Effect", 2015, Angrist and Pischke.
3. Stata, A six-month student copy of the Stata/BE version is sufficient (link).
4. "The Effect: An Introduction to Research Design and Causality", Nick Huntington-Klein.

Instructional Technologies: Statistical Software
The class will use the Stata statistical software package. Stata is a fast and versatile program written by economists, making it more intuitive for people in our field. Stata is the program of choice for applied microeconomists.

You can purchase a six-month student copy of Stata for $48 at the following link:
https://www.stata.com/order/new/edu/profplus/student-pricing/

Stata is a required resource. This software will not run on Chromebooks. The details of computer requirements for Stata can be found here:
https://www.stata.com/products/compatible-operating-systems/

I have been programming in Stata for years, so if you have any questions, please let me know. One note – to help you with your programming, I need to see the code. Therefore, copy the code you are working on and email me the text. Too many people come to my office and say, "I typed what was in the handout, and it did not work," then do not show me their code. I need to see what you are doing before I can help. That is why I strongly encourage visiting my office during office hours.

Requirements (and grading weight)

A. Participation (10%)

Students should be prepared to contribute to class discussions (by answering questions and/or asking clarifying questions), going to office hours, pointing out mistakes on the board, presenting homework that appears on the slides, and providing feedback to peers during group presentations. Throughout the semester, I will provide the class with some news media that is relevant to the topics of discussion. You can earn participation points by adding to the conversation about the piece and relating it to what you have learned in class.

B. In-Class Exercises (10%)

Throughout the course, I will assign in-class exercises to be worked on individually or in groups. In-class assignments will be graded on completion. At the end of the semester, a portion of your grade will be determined based on how many of the in-class assignments you complete (total in-class assignments to be determined, but there will be around 10). To deal with unplanned absences, you will be allowed to miss up to two of the assignments without impacting your grade.

C. Homework Assignments, aka Problem Sets (20%)

Homework will involve solving problems and answering questions related to the material we covered in class, as well as performing econometric analysis in Stata. Some questions will require a handwritten answer.

You may use the textbook, personal notes, or other reference material to do the homework. You may and are encouraged to work with your classmates. However, everyone must turn in their own work, perform their own Stata analysis, and write the answers in their own words.

If homework is identical to a classmate's, you may not receive credit for the answer. Please note that all violations will be reported to the Academic Integrity Office in accordance with university policy.

Assignments and due dates will be posted in Canvas. Homework is due by 11:59 p.m. on the specified date. You can submit a physical copy to me in class or upload it to Canvas. For Stata portions of the homework, you must submit your code and output.

10 points will be taken off for each day past the due date that the homework is turned in. Once the homework solutions are posted, you cannot receive above a 40%.

In total, there are 5 problem sets. I will drop your lowest-scoring homework, so only 4 will contribute to your final grade.

Number	Posting date*	Due Dates*	Topics
Problem Set #1	January 27th	February 3rd	Intro to Course; Probability and Statistics Review
Problem Set #2	February 17th	February 24th	Principles of coding; Economic data; Origins of Economic Data; Labs 1, 2, and 3
Problem Set #3	March 5th	March 17th	Big Data, AI, and Machine Learning; Randomized Control Trials; Lab 5
Problem Set #4	March 31st	April 7th	Motivating Regressions; Regression Estimation; OLS Interpretations; Evaluating OLS; Lab 6
Problem Set #5	April 21st	May 5th	Hypothesis Testing Part; Choosing Independent Variables; Choosing Functional Form; Interaction Terms;

			Lab 7
***Tentative dates**			

D. Midterms (20% each, i.e., total 40%)

There will be two in-class, closed-book midterm exams throughout the semester. For midterms, you are allowed to bring in a single-sided 8.5 x 11-inch sheet of paper with anything on it legible to your naked eye. Stata coding will not be required for midterms, i.e., I will not ask for coding on a computer. Both midterms will take place during the regularly scheduled class time in our usual classroom. The first midterm will be on Thursday, February 26th, and the second on Thursday, April 2nd. There are no makeups for midterms, unless you have a documented emergency (e.g., serious injury, death of a family member).

Exam Corrections

I understand that everyone has an off day and that exams don't always reflect the effort put into learning the material. If you are worried about the grade on a midterm, all students are welcome to submit exam corrections. The process is outlined below:

- You have one week[2] to redo the (same) entire exam.
- You must fully write out how to solve the problem (step-by-step) neatly.
- Students cannot ask for my assistance.
- Students can earn up to 25% of the points back (perfect redo = 25%).
  E.g.: If you got 80%, you could earn up to 25% of the 100%-80%=20% back, i.e., up to +5%, which means that your final score would be 80%+5%=85%.
  E.g.: If you got 60%, you could earn up to 25% of the 100%-60%=40% back, i.e., up to +10%, which means that your final score would be 60%+10%=70%.

E. Group Research Project (20%)

At the end of the second week of class, you will be grouped for a research project. The project will require your group to develop an economic research question, find data to answer the question, and perform the statistical analysis necessary to answer the question.

There will be six components of the project that will be graded.

1. Project Proposal (10%) – Each group will submit three potential research questions they wish to answer for their project (open-ended). This assignment requires students to explain
   a. why the research questions are interesting,
   b. what potential data sources they can use,
   c. and how the question is related to economics.

   Due date: February 06, 2026.

[2] From when exams are handed back.

2. Data Description and Summary Statistics (10%) – Each group will submit a 2–3-page document describing the dataset they have chosen (i.e., structure, number of observations, summary statistics of key variables of interest, etc.).
   Due date: February 27, 2026.

3. Regression Specification and Preliminary Results (15%) – Each group will submit a 2-3-page document describing the chosen regression model to answer the research question, as well as the preliminary results.
   Due date: April 03, 2026.

4. Class Presentation (25%) – Each group will present their findings from their research project during a 12-minute presentation. You are required to present background information, your research question, how you answer the question (the regression model), the data you used, your results, and their implications.
   Due date: April 23, 2026.

5. Final Report (30%) – Each group is required to hand in a 7–10-page report that provides background information, your research question, a discussion of your model, data used, presentation of your results, their implications, as well as their limitations. Students are expected to incorporate the feedback from the class presentation.
   Due date: May 05, 2026.

6. Participation (10%) – Each group member will fill out a report that grades each member's participation. Each member will receive the average grade assigned by their group members.
   Due date: May 08, 2026.

Weights and due dates summary table:

Project Proposal (10%)	February 06
Data Description and Summary Statistics (10%)	February 27
Regression Specification and Preliminary Results (15%)	April 03
Class Presentation (25%)	April 23
Final Report (30%)	May 05
Participation (10%)	May 08

A grading rubric is provided during the semester.

F. Optional Final Exam (can replace a midterm)

The optional final exam will take place on Wednesday, May 13th, at 1:05 p.m. in our usual classroom unless you are informed otherwise. This final exam is optional, and your grade on it will replace your lowest midterm score.

You are allowed to bring two 8.5 x 11-inch sheets of paper with anything legible to your naked eye.

You MUST take the exam at the scheduled date and time. If you happen to have too many exams in a single day (i.e., 3 exams in 24 hours) and wish to reschedule the final exam, you must contact the appropriate VT channel (presumably the Office of the Registrar) and obtain permission from the Dean's office (Note: there is a deadline for this). If you miss the final exam and do not have an excuse from the Dean, your midterm scores will be unaffected. There will be no makeup for the final exam, unless you have a documented emergency (e.g., serious injury, death of a family member).

In summary, final grades for the course will be based on the following:

Participation	10%
In-Class Exercises	10%
Homework Assignments	20%
Midterms	20% + 20% = 40%
Group Research Project	20%
Optional Final	Up to 20%

Letter grades will be assigned as follows:

A	93-100	C+	77-79.99
A-	90-92.99	C	73-76.99
B+	87-89.99	C-	70-72.99
B	83-86.99	D	60-69.99
B-	80-82.99	F	0-59.99

Grading Questions or Disputes

I want you to understand how you are evaluated in this class. If you have any questions about your grade or feedback on coursework, please email me to schedule a time to meet. I kindly ask that you wait at least 24 hours after work has been returned before requesting an appointment. This is to give you time to more thoroughly consider the questions you have about your work.

I will calculate the final course grade by using the grades with one or two decimals; the final course grade will not be rounded to the nearest integer. For example, 69.4% will not be rounded to 70, 69.5% will not be rounded to 70, and 69.9% will not be rounded to 70.

Attendance

Attendance is not mandatory, but as mentioned in the **Requirements** part, two components fundamentally rely on attendance: Participation (10%) and In-Class Exercises (10%).

Use of Personal Technology

No electronic devices (laptops/smartphones) are allowed during normal class. Tablets/iPads to take notes are allowed.
Electronic devices (laptops) are required at the Satata Laboratories.
While you may think you are only distracting yourself, your technology use can distract others and impact their learning. Please make sure your phones are set to silent to avoid unnecessary interruptions. I do realize there may be times when you need to take an urgent phone call. In those situations, please quietly excuse yourself from the classroom before answering your phone.

Canvas

Our website within Canvas (https://canvas.vt.edu/) will have the slides, class assignments, information, and more. Please check the website regularly for class updates.

Instructor Office Hours and Attention via Email

I encourage students to refrain from contacting me via email with questions related to the course material. Instead, I strongly recommend visiting my office for personalized assistance with your questions and doubts.

My office hours are designed to be accessible without the need for appointments, making them "open" for anyone who requires assistance. However, please keep in mind that you may occasionally need to share the time slot with others or wait briefly if I am already engaged in a discussion with other students.

I highly encourage you to take an initiative-taking approach by attending office hours if you find yourself in need of clarification or struggling with any aspect of the material. Your initiative-taking engagement ensures that you receive the necessary support precisely when you need it.

I typically respond to emails twice a week:

I respond to those emails that were sent on:	On:
Saturday, Sunday, Monday, and Tuesday	Tuesday
Wednesday, Thursday, and Friday	Friday

Any questions emailed to me concerning course material will be answered via email and could be shared with all students in the class anonymously.

COVID, Flu, and Sickness Policy

In the case of COVID or other contagious diseases, we will follow the current guidance from the Virginia Department of Health. The Virginia Department of Health recommends staying home when exhibiting signs of respiratory illness and returning to school when symptoms have improved for at least 24 hours, and there is no sign of fever. Additionally, it is recommended that upon returning to school or work, you take precautions for five days to prevent the spread of illness to others. If you do get sick, please let me know. I will make reasonable accommodations to make up any material. This policy may be updated as recommendations change.

Use of Generative AI

While the use of AI tools, including ChatGPT, Grok, DeepSeek, Gemini, and other AI-powered writing or research assistive technologies, may be tempting, it is important to consider the ethical implications of their use.

In this course, students are encouraged to consult with the instructor before using any such tools. If approved, the use of AI tools must be properly cited and acknowledged in all submissions. Failure to follow these guidelines may result in a grade penalty or other disciplinary action as determined by the instructor.

Academic Honesty

One of the goals of this course is to prepare you to do well in advanced economics courses, so you must develop a solid understanding of the material. To that end, while collaboration on problem sets is encouraged in this course, copying is not. Students are expected to understand and abide by the University's honor code (https://honorsystem.vt.edu/).

Withdrawal

Sometimes students may need to withdraw from a class because of personal or academic reasons. If your performance in the class is not where you want it to be or if you are experiencing difficulties, please don't hesitate to contact me before withdrawing.
March 02 is the last day to drop individual courses. March 31 is the last day to resign.

Undergraduate Academic Integrity Statement

The Undergraduate Honor Code pledge that each member of the university community agrees to abide by states:

"As a Hokie, I will conduct myself with honor and integrity at all times. I will not lie, cheat, or steal, nor will I accept the actions of those who do."

Students enrolled in this course are responsible for abiding by the undergraduate Honor Code on each assignment and/or exam. A student who has doubts about how the Honor Code applies to any assignment is responsible for obtaining specific guidance from the course instructor before submitting the assignment for evaluation. Students are strongly discouraged from misusing unauthorized online sources, including generative artificial intelligence outlets. Students are strongly encouraged to consult their faculty members regarding the use of such outside materials, as the misuse of these sources may constitute a violation of the Honor Code. Ignorance of the rules does not exclude any member of the University community from the requirements and expectations of the Honor Code

Academic Accommodations
Virginia Tech welcomes students with disabilities into the University's educational programs. The University promotes efforts to provide equal access and a culture of inclusion without altering the essential elements of coursework. If you anticipate or experience academic barriers that may be due to disability, including but not limited to ADHD, chronic or temporary medical conditions, deaf or hard of hearing, learning disability, mental health, or vision impairment, please contact the Services for Students with Disabilities (SSD) office (540-231-3788, ssd@vt.edu, or visit ssd.vt.edu). If you have an SSD accommodation letter, please meet with me privately during office hours as early in the semester as possible to deliver your letter and discuss your accommodations. You must give me reasonable notice to implement your accommodations, which is generally 5 business days and 10 business days for final exams.

If there are aspects of this course that prevent you from learning or exclude you, please let me know as soon as possible. Together, we'll develop strategies to meet both your needs and the requirements of the course.

Basic Needs Statement
For any student who has difficulty affording groceries, accessing sufficient food to eat every day, or who lacks a safe and stable place to live, and if you believe this may affect your performance in this course, you are urged to contact the Dean of Students office for support at 540-231-3787 or complete an interest form to participate in The Market at Virginia Tech.

The Dean of Students, through The Market at Virginia Tech, offers food options and other resources. There is also a Student Emergency Fund program. If you are comfortable in doing so, please notify your departmental advisor or me of your situation. This will enable them to provide any resources they have access to.

Lauren's Promise: I will listen and believe you if someone is threatening or harassing you.

Lauren McCluskey, a 21-year-old student at the University of Utah and the daughter of economist Jill McCluskey, was murdered on October 22nd, 2018, by a man she had briefly dated. I am committed to doing what I can to make sure this does not happen again.

If you are in immediate danger, call 911. If you are experiencing sexual assault, domestic violence, stalking, or harassment, you can report it to me. You should be aware that I am a mandatory reporter, which means that I am required to report these instances to the university's Title IX Coordinator or Deputy Title IX Coordinator to investigate. **You can also learn about your options (both on and off-campus, as well as confidential or not) at https://oea.vt.edu/title-ix-vawa.html**. If you need support for your mental or emotional health for any reason, you can talk to me. You can also find helpful resources at **https://ucc.vt.edu/**.

Weather Policy
If the University is officially closed on an exam or assignment due date, the exam or the due date of the assignment will be moved to the next scheduled class date.

Mental Health Wellness
Ensuring the well-being of our students, both academically and mentally, is a top priority at Virginia Tech and for me. If you find yourself feeling overwhelmed by academic demands, grappling with mental health challenges, or concerned about the welfare of a friend, please know that there are resources available to support you:

- Cook Counseling: Call 540-231-6557 to schedule an appointment and/or 24/7 crisis support or go online at ucc.vt.edu for more info.
- Dean of Students Office: Call 540-231-3787 for general advice. Call 540-231-6411 for after-hours crisis or go online at dos.vt.edu for more info.
- Hokie Wellness: Go online to hokiewellness.vt.edu for more info about health and wellness workshops and consultations.

You can find a full listing of campus resources at Well-being at Virginia Tech.

Tentative Course Outline (Subject to Change)

Week	Date	Tuesday	Thursday
Week I	Jan. 20th & Jan. 22nd	Inroduction to Course	1.1 Probability and Statistics Review
Week II	Jan. 27th & Jan.29th	1.2 Probability and Statistics Review	2. Principles of coding
Week III	Feb. 3rd & Feb 5th	Lab 1: Installation and introduction to Stata	3. Economic data
Week IV	Feb. 10th & Feb 12th	Lab 2: Data manipulation	4. Origins of the economic data
Week V	Feb. 17th & Feb 19th	Lab 3: Data analysis	5. Big Data, AI, and Machine Learning
Week VI	Feb. 24th & Feb 26th	Lab 4: Frames and graphs	**Midterm 1**
Week VII	Mar. 3rd & Mar. 5th	6. Randomized Control Trials	Lab 5: RCTs
	Mar. 10th & Mar. 12th	Spring Break	Spring Break
Week VIII	Mar. 17th & Mar. 19th	7.1 Motivating Regressions	7.2 Regression Estimation
Week IX	Mar. 24th & Mar. 26th	8.1 OLS Interpretations	8.2 Evaluating OLS
Week X	Mar. 31th & Apr. 2nd	Lab 6: Regression; OLS	**Midterm 2**
Week XI	Apr. 7th & Apr. 9th	9.1 Hypothesis Testing Part 1	9.2 Hypothesis Testing Part 2
Week XII	Apr. 14th & Apr. 16th	10. Choosing Independent Variables & Functional Form	11. Interaction Terms
Week XIII	Apr. 21th & Apr. 23th	Lab 7: Hypothesis Testing; Interaction Terms	**Research Project Presentations**
Week XIV	Apr. 28th & Apr. 30th	12. Multicollinearity/Serial Correlation	13. Heteroskedasticity and Endogeniety
Week XV	May 5th & May 7th	Final Exam Review	Reading Day: No Class
Potential topics conditional on time		Fixed Effects	Difference-in-Differences
		Instrumental Variables	Regression Discontinuity